

Structural Redesign of the Steering System for Team Falcon E Formula Student Car

Reason for Redesign — Previous Steering Column Bearing Mount

The primary motivation for redesigning the steering system was a critical structural weakness identified in the previous season's steering column mount. The yellow bracket assembly, visible in Figure 1, was responsible for housing the bearing collars of both the steering column and steering shaft. This mount was configured as a cantilevered beam — attached to the chassis at one end only, with no secondary support point at the opposite end. Under normal driving conditions, when steering effort was applied by the driver, the cantilevered yellow bracket experienced significant bending loads at its root. This caused the mount to flex noticeably, resulting in excessive steering play and a loss of precision in driver input transmission. The flexing also increased the perceived steering effort required, reducing driver confidence and overall vehicle responsiveness. As a result, a complete redesign of the steering mount was undertaken to eliminate the cantilever condition and provide a structurally rigid, well-supported bearing housing capable of withstanding the steering loads without deformation.

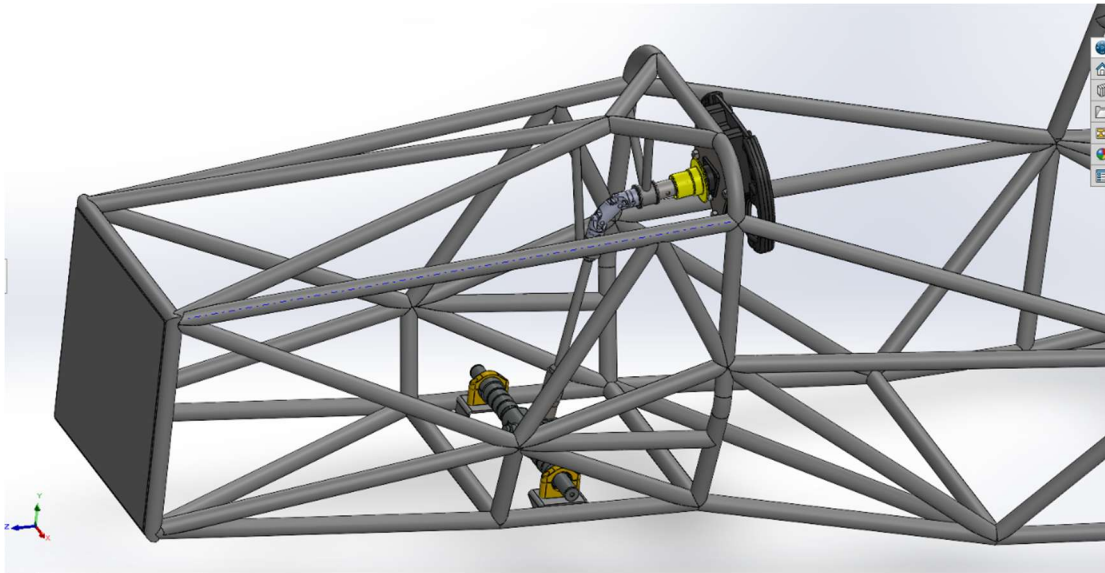


Figure 1 -Previous Steering Column Bearing Mount

Redesign of the Steering Column Bearing Mount

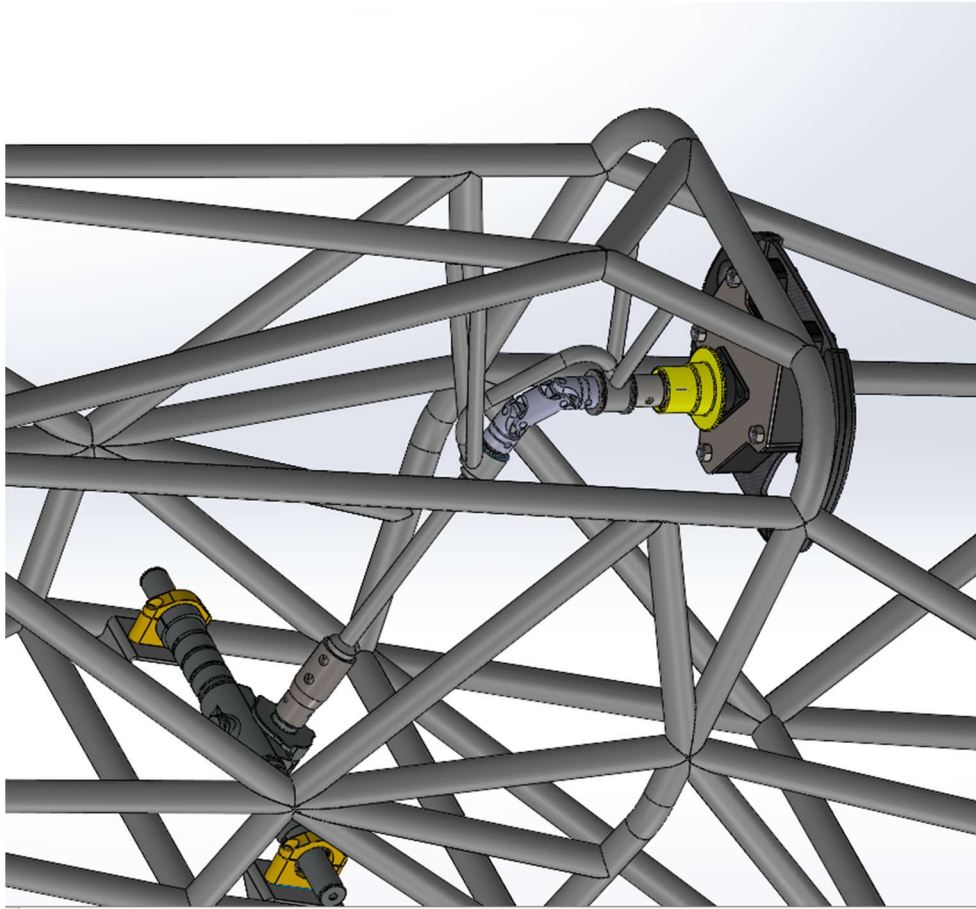


Figure 2 -Improved steering mount

To address the structural weakness of the previous cantilevered configuration, the steering column bearing mount was completely redesigned. The single cantilevered mount was replaced with a triangular mount structure, which distributes the steering loads across multiple contact points rather than concentrating them at a single root. This triangular configuration significantly reduces the bending moment experienced by the mount under applied steering effort, eliminating the flexing observed in the previous design. Additionally, the two mounting points were interconnected using a bent pipe, which further improves the structural rigidity of the overall assembly by tying the mounts together and preventing independent deflection under load. The combination of the triangular mount geometry and the connecting bent pipe resulted in a considerably stiffer and more precise steering system, with noticeably reduced steering play and improved driver feedback.

Design of an Alignment Jig for Bearing Collar Positioning

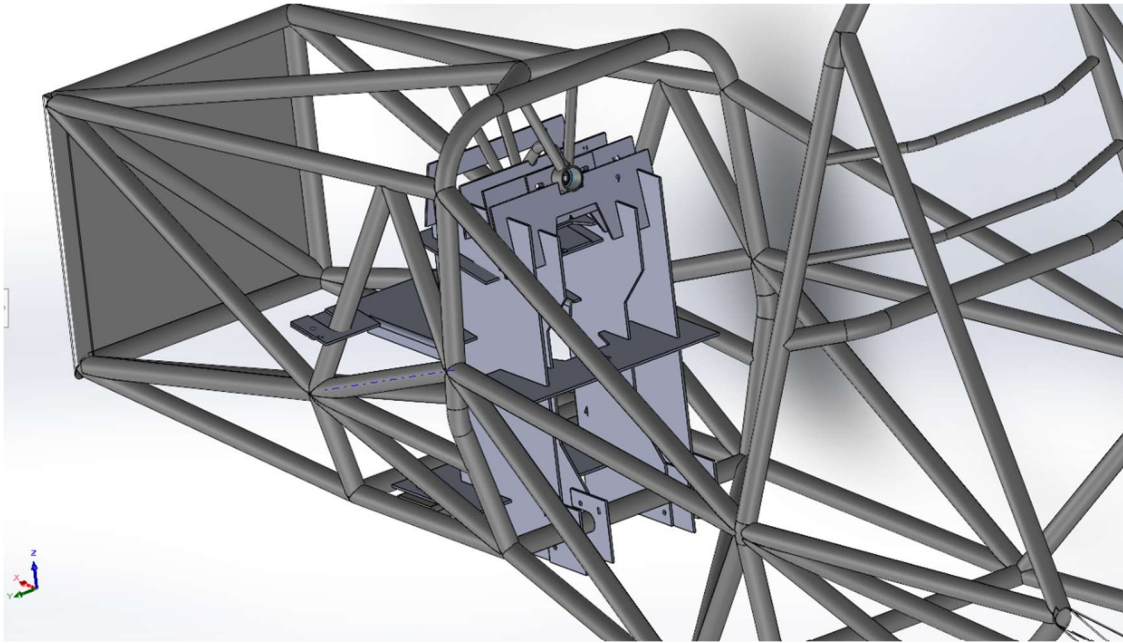


Figure 3 -Alignment Jig for bearing positioning

To ensure precise alignment of both steering column bearing collars during assembly, a dedicated alignment jig was designed. Accurate alignment of the bearing collars is critical, as any angular or axial misalignment between the two collars introduces binding in the steering shaft, increases steering effort, and accelerates bearing wear. The jig was designed to hold both bearing collars in their exact coaxial positions simultaneously, ensuring that the mounted collars are perfectly in line with the steering shaft axis before and during the welding and assembly process. This approach eliminates the risk of misalignment caused by manual positioning and ensures repeatability if multiple units are to be produced.

Manufacturing Phase - Assembly and Welding Using the Alignment Jig

Following the completion of the design phase, the steering column bearing mount was fabricated and assembled using the dedicated alignment jig. Prior to welding, the jig was precisely aligned using a laser alignment tool, ensuring that both bearing collars were held in perfect coaxial alignment with the steering shaft axis. Once the laser alignment was confirmed, the components were secured within the jig and the mount was welded in position. The use of laser alignment eliminated any manual positioning error and guaranteed that the final welded assembly maintained the exact bearing collar positions defined in the CAD model. This process ensured a binding-free, low-friction steering shaft installation with minimal steering play in the finished assembly.



Figure 4 -Alignment using laser



Figure 5 - Welding Process



Figure 6 -Steering system after fabrication

